

Breaking the yield–selectivity trade-off in polystyrene waste valorization via tandem depolymerization and hydrogenolysis

Received: 18 November 2024

Accepted: 16 October 2025

Published online: 08 December 2025



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Converting plastic waste into valuable products mitigates plastic pollution and lowers the carbon footprint of naphtha-derived aromatics. However, the difficulties of precisely controlling complex multiphase systems and the catalyst inefficiencies hinder process viability. Here we report a vapour-phase hydrogenolysis strategy catalysed by Ru single atoms on Co_3O_4 ($\text{Ru}_{\text{SA}}/\text{Co}_3\text{O}_4$), decoupling depolymerization from hydrogenolysis to overcome the toluene yield–selectivity trade-off. In a pressurized dual-stage fixed-bed reactor, polystyrene undergoes hydropyrolysis at 475 °C, followed by vapour-phase hydrogenolysis at 275 °C (0.4 MPa H_2 , 2.4 s), yielding toluene with 99% selectivity, 83.5 wt% yield and 1,320 mmol $\text{g}_{\text{cat}}^{-1} \text{h}^{-1}$ rate. The $\text{Ru}_{\text{SA}}/\text{Co}_3\text{O}_4$ catalyst demonstrates excellent stability, maintaining >99% conversion and selectivity during 100 h continuous operation (turnover number 24,747), and effectively processes diverse real-world polystyrene wastes. Life-cycle assessment shows a 53% carbon footprint reduction over fossil-based methods, while techno-economic analysis estimates a competitive minimum selling price of US\$0.61 kg^{-1} , below the US\$1 kg^{-1} industry benchmark.

Aromatic hydrocarbons such as benzene, toluene and xylenes (BTX), which underpin the production of chemicals and fuels^{1–4}, are ~70% derived from fossil-based naphtha reforming and account for ~7% of global greenhouse gas emissions (Fig. 1a). Plastic manufacturing, another major source of carbon and resource pressure, is projected to exceed 1,100 Mt by 2050, further compounding global climate impacts⁵. Yet only 9% of plastic waste is currently recycled^{6,7}. While mechanical recycling is viable for polyesters, it remains largely ineffective for polyolefins^{8–11}. By contrast, chemical conversion of plastic waste into aromatics offers a route to reduce carbon emissions and fossil feedstock reliance, aligning with carbon neutrality targets^{12–14}.

Among the difficult-to-recycle plastic fractions, polystyrene (PS) has attracted particular interest for catalytic valorization into aromatics. Recent approaches target either functionalized aromatics via oxidative or photochemical methods, or bulk BTX compounds through hydrogenolysis or catalytic pyrolysis^{15–18}. Notably, toluene stands out owing to its central role in the petrochemical industry, with global demand projected to rise from 37 Mt to 77 Mt by 2035¹⁹. However, BTX-directed methods often result in broad product distributions and limited ability to tune selectivity, with toluene yields typically below 12 wt% and selectivity under 13% (refs. 20–22) (Fig. 1b). This limitation arises from the complexity of multiphase (gas–solid–liquid)

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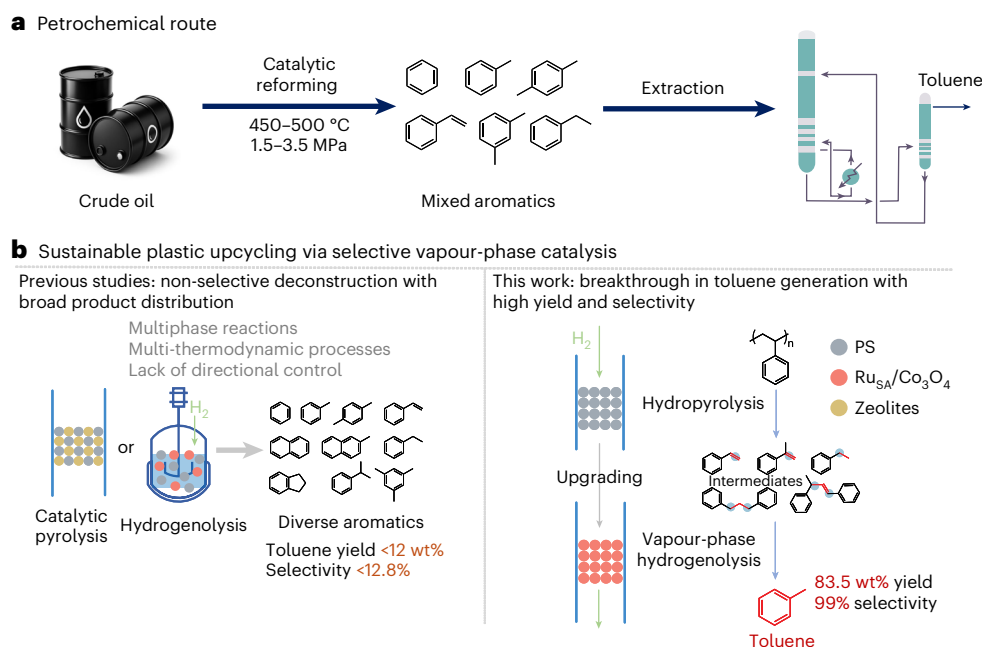


Fig. 1 | Pathways for toluene production from crude oil and PS waste.

a, Petrochemical route for mixed aromatic hydrocarbon production. Conventional catalytic reforming process for toluene production from crude oil, operating at 450–500 °C and 1.5–3.5 MPa, yielding a mixture of aromatic

compounds followed by extraction to obtain toluene. **b**, Selective vapour-phase catalytic upgrading of plastic waste into high-purity toluene. Comparison between previous approaches yielding diverse aromatic products and the selective hydrogenolysis strategy in this work.

reaction environments, where a thermodynamic mismatch between depolymerization and hydrogenation hampers selective C–C bond cleavage^{23–25}. A representative case is hydrogenolysis in high-pressure autoclaves, where pressure–temperature feedback destabilizes the reaction environment. Beyond process-related constraints, catalyst design also plays a critical role in product selectivity. Zeolite catalysts (for example, Zeolite Socony Mobil, ZSM-5) are widely used in catalytic pyrolysis of plastic waste but typically generate C₆–C₁₃ aromatics with mixed monocyclic and polycyclic structures^{26–29} (Fig. 1b). Advances in process control and catalytic material engineering are essential for selective and efficient plastic waste valorization.

Here we report a tandem catalytic strategy combining hydropyrolysis and vapour-phase hydrogenolysis to selectively convert PS waste into toluene. The first stage thermally depolymerizes PS into aromatic intermediates, followed by gas-phase hydrogenolysis that directs C–C bond cleavage and enhances toluene selectivity over a solid catalyst. This tandem design adopts a gas–solid configuration that decouples depolymerization from hydrogenolysis, enhancing reaction control and scalability (Fig. 1b). To boost toluene selectivity, we developed a Ru single-atom catalyst on Co₃O₄, whose atom-level control affords high activity and selectivity^{30–36}. In particular, Ru_{SA}/Co₃O₄ was designed to selectively cleave C–C bonds in key intermediates, steering the reaction towards the formation of toluene. Under reaction conditions of 0.4 MPa and 275 °C in the second-stage hydrogenolysis reactor, the Ru_{SA}/Co₃O₄ demonstrated a toluene selectivity of 99%, with a yield of 83.5 wt% and a formation rate of 1,320 mmol g_{cat}^{–1} h^{–1}. Such selectivity streamlines product separation and improves energy efficiency, facilitating scalable plastic waste valorization.

Catalyst design and characterization

The Ru_{SA}/Co₃O₄ catalyst was prepared via hydrothermal synthesis of a cobalt carbonate precursor, followed by impregnation and calcination^{5,37–39}. Energy-dispersive X-ray spectroscopy (Fig. 2a) confirmed the uniform distribution of many small Co₃O₄ nanoparticles, approximately 20 nm in size. Scanning electron microscopy images

showed that the synthesized Co₃O₄ exhibited a nanosheet morphology with sizes ranging from 500 nm to 1,000 nm (Supplementary Fig. 1a). Transmission electron microscopy images (Supplementary Fig. 1b,d) further revealed a lattice spacing of 0.467 nm, corresponding to the (111) plane of Co₃O₄. In addition, a weak peak at 284.2 eV in the electron energy loss spectrum was ascribed to the M₄ edge of Ru (Supplementary Fig. 2), while a second peak at 543.5 eV corresponded to the O K-edge. These findings confirm the successful incorporation of Ru atoms into the Co₃O₄ structure and suggest a strong interaction between Ru and oxygen species. Ru nanoparticles on Co₃O₄ (Ru_{NP}/Co₃O₄) were prepared as a control, exhibiting structure and dispersion comparable to the single-atom catalyst (Supplementary Fig. 3).

X-ray diffraction analysis confirmed that the Co₃O₄ phase (powder diffraction file no. 43-1003) remained unchanged with either Ru single atoms or nanoparticles as active sites (Supplementary Figs. 4 and 5 and Supplementary Tables 1 and 2). High-resolution transmission electron microscopy was used to further examine the microstructure. Aberration-corrected high-angle annular dark-field scanning transmission electron microscopy imaging (Fig. 2b) visualized the Ru single atoms, and slight rotation of the (111) plane during imaging enhanced the clarity of the Ru atom distribution. High-contrast spots, highlighted by blue circles in the magnified image (Fig. 2c), represent the single Ru atoms dispersed on the Co₃O₄ nanosheets. No evidence of Ru atom aggregation was observed, confirming successful dispersion.

We characterized the electronic structure of the Ru_{SA}/Co₃O₄ catalyst using X-ray photoelectron spectroscopy to explore the interaction between Ru_{SA}s and the Co₃O₄ support. In the high-resolution Co 2p spectrum (Supplementary Fig. 6), an increase in binding energy for both the 2p_{3/2} and 2p_{1/2} peaks indicated charge transfer between Ru and Co, with Ru oxidizing some Co sites and increasing the Co³⁺ component^{40,41} (Supplementary Table 3). However, no notable differences in the electronic state of Co were observed between Ru_{SA}s and Ru_{NPs}. The high-resolution O 1s spectrum revealed that the single-atom Ru catalyst has a lower defect oxygen content compared with the support, resulting from its coordination with oxygen vacancies in Co₃O₄.

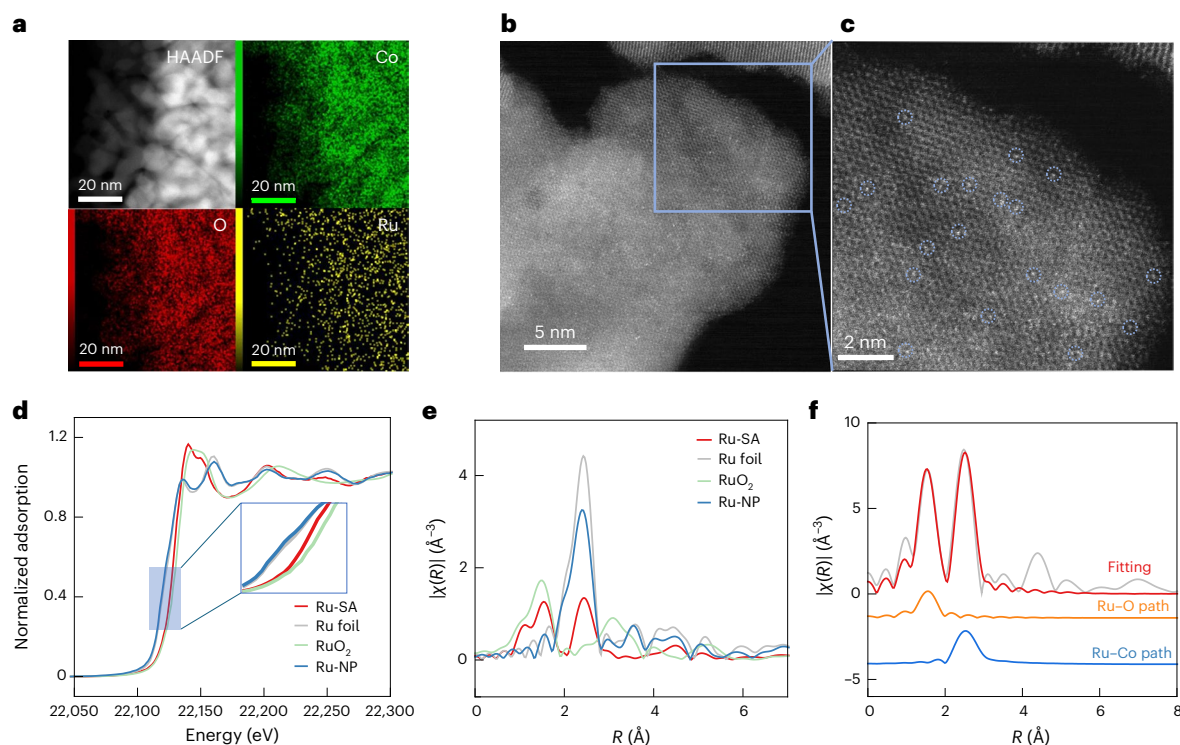


Fig. 2 | Structural and electronic characterization. **a**, Energy-dispersive X-ray spectroscopy mapping images of $\text{Ru}_{\text{SA}}/\text{Co}_3\text{O}_4$ (green represent Co element, red represent O, and yellow represent Ru element). **b**, Atomic-scale structure of $\text{Ru}_{\text{SA}}/\text{Co}_3\text{O}_4$. High-angle annular dark-field (HAADF) scanning transmission electron microscopy image of $\text{Ru}_{\text{SA}}/\text{Co}_3\text{O}_4$. **c**, Enlarged region in **b** with blue circles marking

the locations of Ru atoms. **d**, Normalized XANES spectra of the Ru K-edge. **e**, R -space extended X-ray absorption fine structure spectra of $\text{Ru}_{\text{SA}}/\text{Co}_3\text{O}_4$ and reference materials. **f**, Extended X-ray absorption fine structure fitting results for $\text{Ru}_{\text{SA}}/\text{Co}_3\text{O}_4$. Inset depicting the model used for fitting with two coordination shells.

By contrast, Ru_{NPs} increased surface oxygen defects, because their larger size alters interactions with the support. The Ru $3p$ spectrum revealed that the $3p_{1/2}$ peak for Ru_{SA} s appeared at 463.8 eV, higher than the typical 461 eV for Ru (0), confirming the oxidized state of Ru within the Co_3O_4 substrate.

We used X-ray absorption spectroscopy to examine the oxidation state of Ru, based on the absorption edge position (Fig. 2d). In the Ru_{SA} s catalyst, Ru exhibits a +3 oxidation state, while in Ru_{NPs} , it remains in a zero-oxidation state. The X-ray absorption near edge structure (XANES) spectra reveal that the white line peak and near-edge absorption reflect multiple scattering effects around the central Ru atoms. Comparable to Ru foil, the Ru_{NPs} exhibit a near-edge adsorption peak shape but with diminished intensity, a consequence of their smaller particle size and unsaturated coordination structure. By contrast, the Ru_{SA} s show a near-edge peak distinct from both Ru foil and RuO_2 , indicating a unique Ru coordination environment. XANES-derived insights were validated by FEFF9 simulations using a single-atom Ru model, which reproduced experimental near-edge features⁴¹ (Supplementary Fig. 7a). Partial density of states analysis attributes the higher-energy spectral feature to Ru–Co electronic interactions, underscoring the role of Ru–O–Co coordination in shaping the XANES response (Supplementary Fig. 7b).

A detailed analysis of the single-scattering signal in the extended X-ray absorption fine structure region, processed with K^2 -weighting, reveals coordination differences between Ru_{SA} s and the reference sample (Supplementary Fig. 8). In the high- K region ($K > 8 \text{ \AA}^{-1}$), reduced peak intensity indicates long-range disorder in the Ru atomic arrangement, particularly in the third coordination shell, confirming the absence of Ru-based compound formation and aligning with the design of single-dispersed Ru sites. R -space transformation of the Ru_{SA} s spectrum shows two prominent peaks at approximately 1.5 Å and 2.4 Å (Fig. 2e). The first peak corresponds to Ru–O coordination, while the second is

attributed to long-range Ru–Co interactions, with the Ru–Co distance exceeding that of typical Ru–Ru bonding. On the basis of R -space curve analysis and morphology characterization, we proposed a $\text{Ru}/\text{Co}_3\text{O}_4$ (111) model (Fig. 2f). Fitting this model to the R -space curves using IFEFFIT revealed a coordination number of approximately 5-fold for Ru–O and 9-fold for the Ru–Co second coordination shell, suggesting the presence of Co vacancies in the support (Supplementary Table 4). Wavelet transform analysis (Supplementary Fig. 9), integrating both K -space and R -space data, further supports the proposed structure and curve interpretation.

Vapour-phase hydrogenolysis of model compounds

We investigated the selective vapour-phase hydrogenolysis of styrene, the monomer of PS, to produce toluene using various catalysts. The experiments were conducted in a pressurized dual-stage fixed-bed reactor, where styrene was vaporized in the first stage and subjected to hydrogenolysis in the second stage at 275 °C and 0.4 MPa (Supplementary Figs. 10–12). Among all catalysts investigated, Co_3O_4 achieved a 66.4% conversion of styrene, with ethylbenzene as the major product and only 7.2% toluene (Fig. 3a). This result indicates that Co_3O_4 can promote the hydrogenation of the vinyl group ($-\text{CH}=\text{CH}_2$) in styrene to form ethylbenzene, but has poor activity to cleave the $\text{C}_{\text{sp}3}-\text{C}_{\text{sp}3}$ bond in ethylbenzene to produce toluene.

Introducing Ru nanoparticles on Co_3O_4 ($\text{Ru}_{\text{NP}}/\text{Co}_3\text{O}_4$) substantially enhanced the catalytic performance, increasing styrene conversion to 87.9%. The main products were benzene and ethylbenzene, suggesting that $\text{Ru}_{\text{NP}}/\text{Co}_3\text{O}_4$ promote the cleavage of the $\text{C}_{\text{sp}2}-\text{C}_{\text{sp}3}$ bond between the benzene ring and the ethyl group in ethylbenzene. Remarkably, dispersing ruthenium as single atoms on Co_3O_4 ($\text{Ru}_{\text{SA}}/\text{Co}_3\text{O}_4$) led to complete conversion of styrene and a pronounced shift in selectivity. It

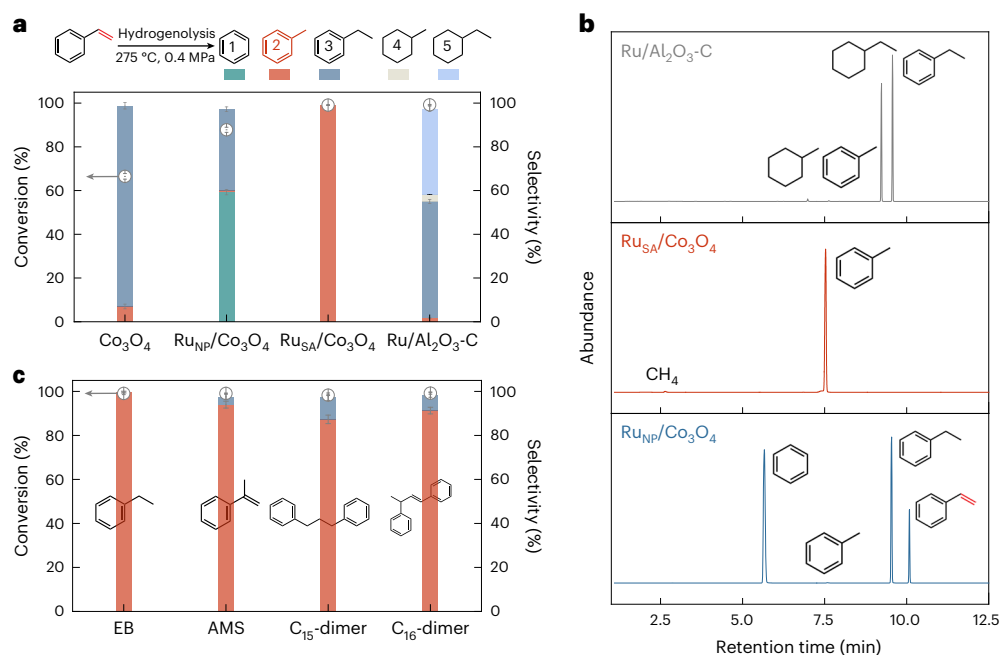


Fig. 3 | Catalytic vapour-phase hydrogenolysis of representative model substrates. a, Conversion and selectivity profiles for styrene hydrogenolysis over different catalysts. **b**, Total ion chromatograms of products obtained from hydrogenolysis of styrene over different catalysts. **c**, Vapour-phase hydrogenolysis performance on additional PS-derived substrates over Ru_{SA}/Co₃O₄ catalyst. Selectivity values represent the distribution of aromatic

hydrocarbon products and exclude gaseous by-products such as methane. The vapour-phase hydrogenolysis was conducted at 275 °C and 0.4 MPa. Catalyst loading was 30 mg, with a catalyst-to-feedstock mass ratio of 10:1. Each data point represents the mean of three independent experiments ($n = 3$), and error bars denote the standard deviation. EB and AMS denote ethylbenzene and α-methylstyrene, respectively.

gave an exceptional selectivity of toluene (>99%) among the aromatic hydrocarbons, demonstrating its high efficiency in transforming the vinyl group of styrene into a methyl group. Chromatograms in Fig. 3b show only a single toluene peak with Ru_{SA}/Co₃O₄, indicating that no other aromatic by-products were detected. This contrasts with other catalysts, which produce multiple peaks corresponding to diverse products such as benzene and ethylbenzene. For comparison, a commercial 5 wt% Ru/Al₂O₃ catalyst also achieved complete styrene conversion, with its catalytic activity contributing notably to hydrogenation of the aromatic ring, resulting in the formation of ethylbenzene and ethylcyclohexane as major products (Fig. 3a).

The selective hydrogenolysis of Ru_{SA}/Co₃O₄ was further evaluated using additional common products from PS depolymerization, including ethylbenzene, methylstyrene and C₁₅/C₁₆ dimers, besides styrene. As shown in Fig. 3c, near-complete conversion of each substrate was achieved, with toluene as the primary product and ethylbenzene as the only minor aromatic by-product. These results underscore the efficiency of Ru_{SA}/Co₃O₄ in facilitating selective C_{sp3}-C_{sp3} bond cleavage between the benzyl group and the alkyl side chains, highlighting its potential for converting PS into toluene.

Tandem depolymerization and hydrogenolysis of PS waste

We further examined the tandem conversion of PS (weight-average molecular weight, $M_w = 192$ kDa; Supplementary Figs. 13–15 and Supplementary Table 5) to toluene in the same reactor. Hydrolysis of PS generated unstable aromatic intermediates (for example, styrene and its dimers), which were stabilized via online vapour-phase hydrogenolysis to selectively produce toluene. This integrated gas–solid process eliminates energy-intensive condensation and reheating steps required in conventional pyrolysis–hydrogenation schemes (Supplementary Fig. 16a), thereby improving energy efficiency and lowering carbon emissions. Enhanced mass transfer and precise reaction control enable better product selectivity, while avoiding multiphase

(gas–liquid–solid) complications improves scalability and tunability compared with traditional systems (Supplementary Fig. 16b).

Optimization with Ru_{SA}/Co₃O₄ identified key parameters affecting toluene yield and selectivity. Increasing the first-stage hydrolysis temperature from 425 °C to 475 °C raised the yield from 56.3 wt% to a peak of 83.5 wt% (Supplementary Fig. 17), with no further improvement at higher temperatures. In the second-stage hydrogenolysis reactor, 200 °C yielded only 40.5 wt% toluene with by-products such as ethylbenzene and ethylcyclohexane (Supplementary Figs. 18 and 19), while 275 °C gave the highest yield. At 350 °C, excessive cracking led to 97.3 wt% methane (Supplementary Figs. 20 and 21). Elevated pressure (1.0 MPa) suppressed toluene formation and favoured cycloalkanes such as methylcyclohexane (90% selectivity; Supplementary Figs. 22–24). Notably, methylcyclohexane serves as both a hydrogen carrier and high-density aviation fuel, underscoring the versatility of this waste-to-value strategy.

At 475 °C hydrolysis, 275 °C vapour-phase hydrogenolysis and 0.4 MPa H₂, PS conversion yielded gas, liquid and solid products (Fig. 4a), with liquid products dominating. Without a catalyst, styrene was the main product (89.7% selectivity), along with toluene (2.7%), ethylbenzene, methylstyrene and C₁₅–C₁₆ aromatics (Fig. 4b and Supplementary Fig. 25). Co₃O₄ primarily produced ethylbenzene (60.3%) via vinyl hydrogenation, while Ru_{NP}/Co₃O₄ favoured benzene (62.3%) and ethylbenzene (31.7%). By contrast, Ru_{SA}/Co₃O₄ achieved near-exclusive toluene selectivity (83.5 wt%). Scale-up tests with liquid-phase product collection gave comparable yields (82.9 wt%), confirming catalytic robustness (Supplementary Figs. 26 and 27). The high specificity arises from selective cleavage of C_{sp3}-C_{sp3} bonds in the vinyl group while preserving C_{sp2}-C_{sp3} bonds. Time-resolved analysis further revealed product evolution dynamics (Supplementary Figs. 28 and 29).

The performance of Ru_{SA}/Co₃O₄ was compared with that of ZSM-5, a widely used catalyst in the catalytic pyrolysis of plastics for producing aromatic hydrocarbons^{11,42}. ZSM-5 yielded aromatic compounds with

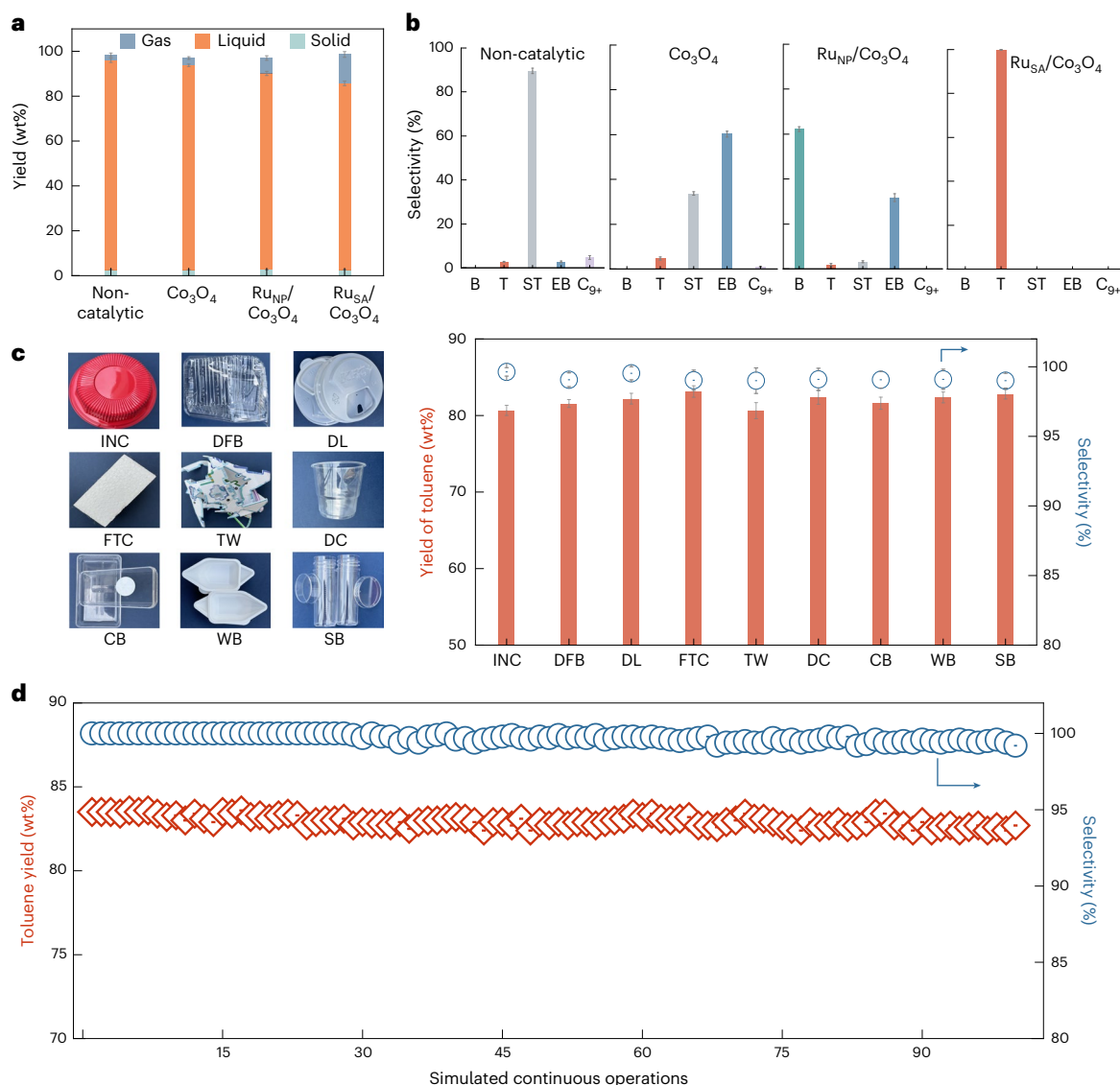


Fig. 4 | Tandem valorization of PS waste into toluene. a, Product distribution across gas, liquid and solid phases in non-catalytic and catalytic processes at a hydrolysis temperature of 475 °C, a hydrogenolysis temperature of 275 °C and a reaction pressure of 0.4 MPa. Each reaction used 30 mg of catalyst with a catalyst-to-PS mass ratio of 10:1. The H_2 flow rate was set at 100 ml min^{-1} , corresponding to a reaction time of 2.4 s under standard conditions. **b**, Product distribution from PS depolymerization. Selectivity towards benzene (B), toluene

(T), styrene (ST), ethylbenzene (EB) and C_9^+ products across different catalysts. **c**, Scalability testing with post-consumer single-use PS wastes, including an instant noodle container (INC), disposable food box (DFB), drink lid (DL), foam takeout container (FTC), toy waste (TW), disposable cup (DC), cake box (CB), weighing boat (WB) and sample bottle (SB). **d**, Stability of $\text{Ru}_{\text{SA}}/\text{Co}_3\text{O}_4$ under simulated continuous PS feed. Each data point represents the mean of three independent experiments ($n = 3$), and error bars denote the standard deviation.

carbon chain lengths ranging from C_6 to C_{12} ; however, it generated only 1.4% toluene while producing 12.2% polycyclic aromatic hydrocarbons⁴³ (Supplementary Fig. 30). The wide product distribution underscores the challenges of conventional aromatization pathways in directing selectivity towards toluene. We also evaluated our results against previous studies on PS hydrogenolysis. Under $\text{Ru}_{\text{SA}}/\text{Co}_3\text{O}_4$ catalysis at 0.4 MPa, with a calculated reaction time of 2.4 s, our process achieved an 8.4-fold increase in yield (83.5 wt% versus 11.9 wt%) and a 7.8-fold improvement in selectivity (99.9% versus 12.8%) compared with the best-reported values obtained under more stringent conditions⁴⁴ (1 MPa for 6 h; Supplementary Tables 6–8).

$\text{Ru}_{\text{SA}}/\text{Co}_3\text{O}_4$ demonstrated scalability across 9 post-consumer single-use PS wastes (Fig. 4c and Supplementary Figs. 31 and 32), yielding ~81 wt% toluene with >99% selectivity and $1,320 \text{ mmol g}_{\text{cat}}^{-1} \text{ h}^{-1}$ formation rate. The instant noodle container sample yielded comparable toluene selectivity despite red dye, suggesting negligible

interference likely owing to dye decomposition at 475 °C hydrolysis and 275 °C hydrogenolysis. For nitrogen-containing acrylonitrile styrene, $\text{Ru}_{\text{SA}}/\text{Co}_3\text{O}_4$ enabled 92.7% toluene selectivity while eliminating toxic by-products such as acrylonitrile (Supplementary Figs. 33 and 34). The system also tolerated halogen-containing PS wastes (expanded polystyrene (EPS), extruded polystyrene (XPS) and mixtures), consistently delivering ~80 wt% liquid yield and >97% toluene selectivity (Supplementary Fig. 35). Notably, real-world mixed PS waste maintained >99% toluene selectivity (Supplementary Fig. 36), highlighting the feedstock flexibility and robustness of this strategy.

To assess the durability and scalability of $\text{Ru}_{\text{SA}}/\text{Co}_3\text{O}_4$, we performed continuous vapour-phase hydrogenolysis using ethylbenzene, a representative PS-derived intermediate less prone to polymerization than styrene. In a fixed-bed reactor (275 °C, 0.4 MPa H_2 , weight hourly space velocity = 2.6 h^{-1}), the catalyst maintained >99% conversion and >99% toluene selectivity for 100 h (Supplementary Fig. 38),

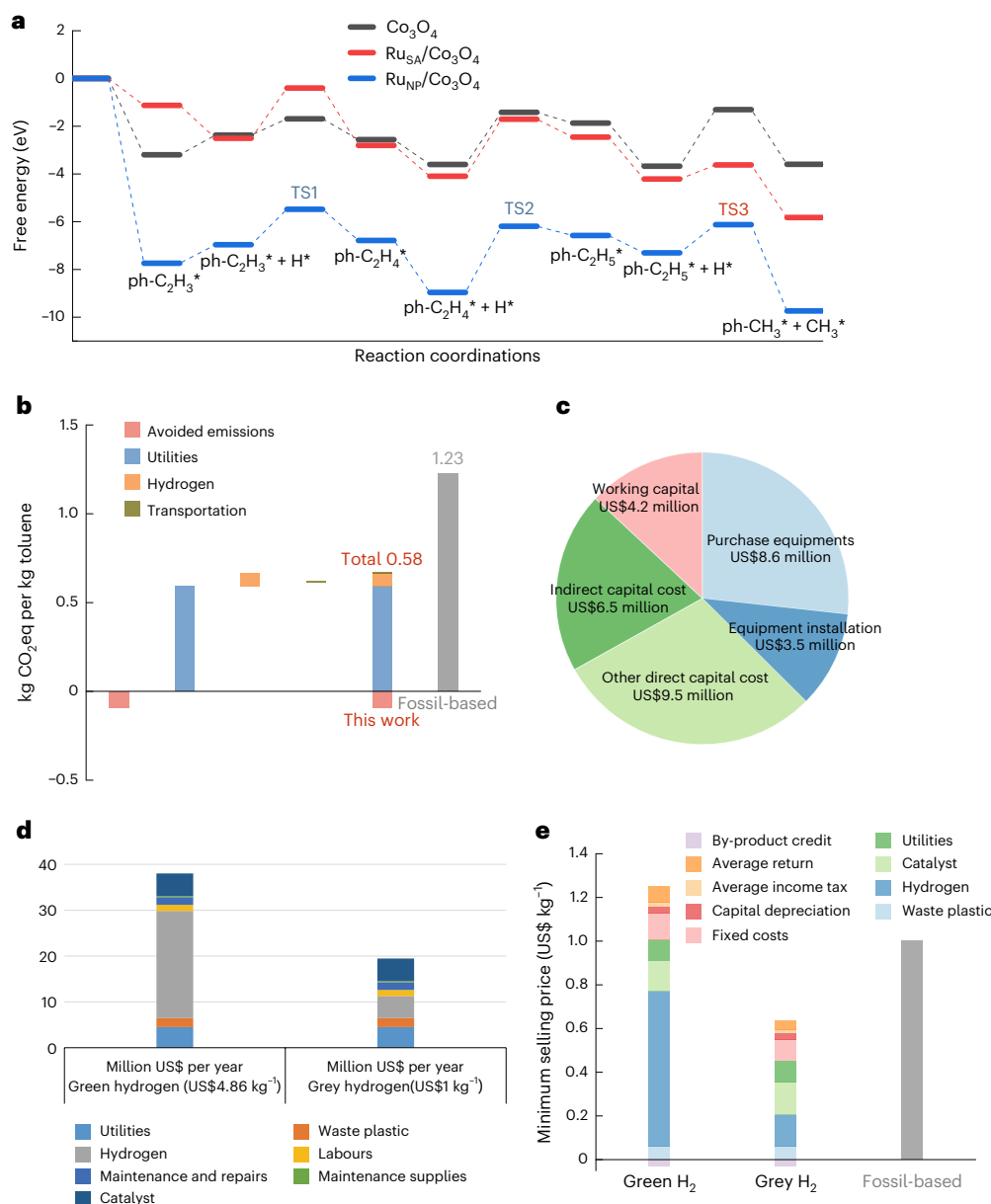


Fig. 5 | DFT mechanistic study, LCA and TEA of sustainable toluene production from PS waste. a, Free energy profiles of reaction pathways for key reaction steps on Co_3O_4 , $\text{Ru}_{\text{SA}}/\text{Co}_3\text{O}_4$ and $\text{Ru}_{\text{NP}}/\text{Co}_3\text{O}_4$ catalysts. **b**, Life-cycle emissions for toluene production. Greenhouse gas emissions per kg of toluene produced, showing reductions achieved in this work compared with fossil-based methods. **c**, Capital cost distribution. Breakdown of capital investment for toluene production from PS, including contributions from working capital, equipment

purchase, installation, indirect costs and other direct costs. **d**, Operating cost comparison. Annual operating costs for toluene production using green hydrogen (US\$4.86 kg^{-1}) and grey hydrogen (US\$1 kg^{-1}), detailing costs for utilities, hydrogen, waste plastic, catalyst and maintenance. **e**, Minimum selling price of toluene. Comparison of the minimum selling price of toluene produced with green hydrogen, grey hydrogen and fossil-based processes, considering costs and by-product credits.

achieving a high turnover number (TON) of 24,747. However, the low melting points of plastics hinder continuous solid feeding, as radiant heating in fixed-bed systems often causes premature softening and feeder blockage. Consequently, most studies assess catalyst stability via regeneration cycles rather than continuous polymer feeding^{28,44} (Supplementary Table 9). To address this challenge and emulate practical operations, we adopted an intermittent feeding strategy: PS was periodically introduced into a first-stage reactor, while the $\text{Ru}_{\text{SA}}/\text{Co}_3\text{O}_4$ catalyst remained stationary in the second-stage fixed-bed reactor for vapour-phase hydrogenolysis. The catalyst consistently delivered an 83 wt% yield of toluene with >99% selectivity across 100 feed cycles (Fig. 4d), indicating long-term stability and compatibility with real plastic waste. Raman and thermogravimetric analyses (TGA) analyses

revealed negligible coking, and structural characterizations confirmed the intact Ru single-atom configuration (Supplementary Figs. 39–41).

Reaction mechanism and sustainability evaluation

To rationalize the mechanistic origin of the high styrene conversion and toluene selectivity observed over $\text{Ru}_{\text{SA}}/\text{Co}_3\text{O}_4$, we conducted density functional theory (DFT) calculations. The catalytic performance was found to be closely tied to the electronic properties of Ru species. Compared with Ru nanoparticles, Ru single atoms on Co_3O_4 exhibit a notably upshifted d-band centre (−1.3 eV versus −1.9 eV), as revealed by d-band centre theory and Fermi-level analysis (Supplementary Figs. 42–44). Hybridization among Ru 4d, Co 3d and O 2p orbitals in $\text{Ru}_{\text{SA}}/\text{Co}_3\text{O}_4$

suggests that coordination influences the electronic environment around active Ru sites, resulting in distinct adsorption behaviours compared with Ru_{NP}/Co₃O₄. The electron density difference revealed positive formal charges between Ru and its coordinated Co atoms, indicating a tendency for electron transfer from Ru to Co. This electron migration enhances the activity by facilitating the adsorption and activation of reactants on the active sites.

We evaluated the Gibbs free energies for four transition states involved in styrene hydrogenolysis. Ru_{SA}/Co₃O₄ exhibits a lower energy barrier of 2.38 eV in the hydrogenation of the vinyl group in styrene (ph-C₂H₄* + H*, TS2; Fig. 5a), compared with 2.78 eV for Ru_{NP}/Co₃O₄. This indicates that Ru_{SA}/Co₃O₄ has a lower energy barrier in the rate-determining step of vinyl group reduction. Following the hydrogenation of styrene to ethylbenzene, the adsorbed ph-C₂H₅* species at the catalyst interface undergoes C–C cleavage via hydrogenolysis to form toluene (Supplementary Fig. 44). The Co₃O₄ support exhibited a high energy barrier of 2.37 eV for ethylbenzene hydrogenolysis (ph-C₂H₅* + H* → ph-CH₃* + CH₃*, TS3). This barrier was reduced to 1.18 eV by Ru_{NP}/Co₃O₄ and further decreased to 0.59 eV with Ru_{SA}/Co₃O₄, highlighting the superior activity of single-atom Ru sites for this transformation (Supplementary Tables 10–12).

To elucidate the superior toluene selectivity of single-atom Ru compared with Ru nanoparticles, we calculated the adsorption energies of toluene intermediates and their subsequent reactions with H* (Supplementary Fig. 45). The toluene intermediate exhibited stronger adsorption on Ru nanoparticles (–0.80 eV) than on single-atom Ru (–0.51 eV), suggesting that toluene adsorption is less favourable on the single-atom sites. In addition, we conducted a transition state energy barrier analysis for the hydrogenolysis of toluene to benzene. The energy barrier for the key transition state (ph-CH₃* + H* → ph* + CH₃*, TS4) was markedly lower on Ru nanoparticles (1.32 eV) than on single-atom Ru (3.32 eV), indicating that hydrogenolysis of toluene to benzene is less favourable on single-atom Ru compared with nanoparticles. Overall, Ru nanoparticles exhibit a lower energy barrier for toluene hydrogenolysis and a higher barrier for ethylbenzene hydrogenolysis, leading to the formation of a broader range of products during PS depolymerization and thus reducing toluene selectivity.

The life-cycle assessment (LCA) of the waste-to-toluene production evaluates the environmental footprint and sustainability of this circular transformation by quantifying cradle-to-gate emissions (Supplementary Fig. 46 and Supplementary Tables 13–16). This LCA focuses on the global warming potential (GWP) across a 100-year horizon, yielding a value of 0.58 kg CO₂ equivalence (CO₂-eq) per kg of toluene produced, a 53% reduction compared with the petroleum-based toluene benchmark of 1.23 kg CO₂-eq per kg (Fig. 5b). Utilities were identified as the largest contributor to GWP among key process elements, highlighting the potential of renewable energy integration to improve environmental performance.

Alongside the LCA, a techno-economic analysis (TEA) was conducted to evaluate the feasibility of a dedicated facility in Shanghai, China, capable of processing the full accessible feedstock. Using a discounted cash flow model, the minimum viable selling price was determined (Supplementary Tables 17–19). Capital expenditure (CAPEX) was dominated by pyrolysis and hydrotreating units, with additional costs attributed to engineering and contingencies (Fig. 5c). Operating expenditure (OPEX) was driven by hydrogen under green pricing and by catalyst and utilities under grey pricing (Fig. 5d). Feedstock, labour and maintenance were also notable OPEX components.

Sensitivity analysis of the minimum selling price of plastic-derived toluene revealed strong dependence on hydrogen pricing: the minimum selling price increases to US\$1.22 kg^{–1} with green hydrogen and decreases to US\$0.61 kg^{–1} with grey hydrogen, both remaining competitive compared with the commercial toluene price of US\$1 kg^{–1} (Fig. 5e, Supplementary Fig. 47 and Supplementary Tables 20–22). These scenarios highlight the economic advantage of plastic-derived toluene

over fossil-based alternatives, reinforcing the economic feasibility of this sustainable pathway. Future efforts should prioritize hydrogen supply-chain optimization and process-level energy integration to realize scalable, circular plastic-to-toluene technologies.

Conclusion

This study demonstrates a highly selective vapour-phase strategy for converting PS waste into toluene using a Ru single-atom catalyst on Co₃O₄. The tandem hydropyrolysis–hydrogenolysis process achieves 83.5 wt% yield and >99% selectivity under mild conditions, enabled by atomically dispersed Ru that lowers energy barriers for C–C bond cleavage. The catalyst maintains structural integrity and performance over 100 h of continuous operation. Integrated life-cycle and techno-economic assessments support its environmental and economic viability. However, challenges remain in continuous solid-plastic feeding, energy integration and scalability beyond PS. Future work should address feedstock generality, long-term fouling resistance and system-level energy optimization to accelerate industrial translation.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41565-025-02069-x>.

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Methods

Materials

Cobalt(II) nitrate hexahydrate ($\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$, 98%), sodium carbonate (Na_2CO_3 , 99%), polyvinyl pyrrolidone (PVP, 98%), ethanol, methanol and ethylene glycol (analytical grade) were purchased from China National Pharmaceutical Group or Beijing Tongguang Fine Chemical and used without further purification. Ruthenium precursors, including $\text{RuCl}_3 \cdot 3\text{H}_2\text{O}$ and $\text{Ru}(\text{NO})(\text{NO}_3)_3$ (98%), were supplied by Alfa Aesar. PS ($M_w = 192$ kDa) and acrylonitrile styrene plastics were obtained from Sigma-Aldrich. Post-consumer PS wastes were collected from a recycling centre in Nanjing, China. External gas–chromatography (GC) standards (styrene, toluene, ethylbenzene, benzene and methylcyclohexane) were also purchased from Sigma-Aldrich. Commercial ZSM-5 ($\text{Si}/\text{Al} = 23$, surface area = $425 \text{ m}^2 \text{ g}^{-1}$) was obtained from Alfa Aesar.

Catalyst preparation

The Co_3O_4 precursor was synthesized by adding 12.5 ml of concentrated $\text{NH}_3 \cdot \text{H}_2\text{O}$ (solution A) to 15 ml of ethylene glycol (solution B) under stirring. After 2 min, 3.5 ml of 1 M Na_2CO_3 and 5 ml of 1 M $\text{Co}(\text{NO}_3)_2$ were sequentially introduced. The mixture, which gradually turned violet, was stirred for 20 min, transferred to a 50 ml Teflon-lined autoclave and heated at 170°C for 17 h. The resulting solid was washed with deionized water and ethanol, and then dried at 60°C .

To prepare Ru single-atom catalysts ($\text{Ru}_{\text{SA}}/\text{Co}_3\text{O}_4$), 200 mg of Co_3O_4 precursor and 4.88 mg of $\text{RuCl}_3 \cdot 3\text{H}_2\text{O}$ were dispersed in 20 ml of ethanol, sonicated for 5 min and stirred at 40°C for 12 h. The solid was recovered by centrifugation, washed with ethanol, dried at 60°C under vacuum and calcined in air at 300°C for 2 h.

For Ru nanoparticle (Ru_{NP}) synthesis, 7 ml of $\text{Ru}(\text{NO})(\text{NO}_3)_3$ solution (10 mg ml^{-1}), 224 mg of PVP ($M_w = 30,000$) and 193 ml of ethylene glycol were stirred to form a homogeneous solution, which was then refluxed at 200°C for 3 h in air. The product was collected by centrifugation, washed with ethanol and redispersed in 20 ml of ethanol.

To synthesize $\text{Ru}_{\text{NP}}/\text{Co}_3\text{O}_4$, 200 mg of Co_3O_4 precursor and 1 ml of the Ru_{NP} dispersion were mixed in 20 ml of ethanol, sonicated for 5 min and stirred at 40°C for 12 h. After washing and drying, the sample was calcined in air at 300°C for 2 h.

Catalyst characterization

X-ray powder diffraction was performed on a Rigaku RU-200b diffractometer using $\text{Cu K}\alpha$ radiation ($\lambda = 1.5406 \text{ \AA}$). X-ray photoelectron spectroscopy was conducted on an ULVAC PHI Quantera microscope to analyse surface composition. Transmission electron microscopy images were obtained with a Hitachi H-800 microscope to examine catalyst morphology. High-resolution transmission electron microscopy, energy-dispersive X-ray spectroscopy line scans and mappings were captured using a JEOL JEM-2100F operating at 200 kV. High-angle annular dark-field scanning transmission electron microscopy was performed on a JEOL 200F microscope at 200 keV, equipped with a probe spherical aberration corrector. Inductively coupled plasma optical emission spectroscopy was conducted using a Thermo Fisher iCAP PRO instrument.

X-ray absorption fine structure spectra at the Ru K-edge were collected at the 1W1B beamline of the Beijing Synchrotron Radiation Facility (BSRF), operating at 2.5 GeV with a current of 250 mA. Fluorescence-detected spectra were acquired using a 19-element high-purity Ge detector array, with Mo and Fe filters. For X-ray absorption fine structure measurements, 40 mg of $\text{Ru}_{\text{SA}}/\text{Co}_3\text{O}_4$ or $\text{Ru}_{\text{NP}}/\text{Co}_3\text{O}_4$ was ground and pressed into 8 mm pellets. Reacted catalysts mixed with quartz were prepared by grinding 60 mg of the sample for 10 min and pressing into 8 mm pellets. Reference data for Ru foil, RuO_2 and RuCl_3 were collected in transmission mode at both the 1W1B and BL14W1 beamlines.

Tandem fixed-bed reactor and online GC–MS analysis

A custom-designed two-stage pressurized fixed-bed reactor (RX305OTR, Frontier Lab) was used for the upcycling of PS waste.

The set-up consists of two vertically connected stainless-steel reactors with inert coatings and independent temperature control. The first-stage reactor ($76 \times 125 \times 75 \text{ mm}$, $100\text{--}900^\circ\text{C}$) enables hydrolysis of up to 50 mg PS, which is loaded in a stainless-steel sample cup, introduced via a high-pressure solid sampler. The second-stage reactor ($76 \times 125 \times 160 \text{ mm}$, $100\text{--}700^\circ\text{C}$) contains a quartz tube (3 mm inner diameter, i.d.) packed with catalyst ($30 \pm 0.05 \text{ mg}$) diluted with SiO_2 for vapour-phase hydrogenolysis. Both reactors have an inner diameter of 4.6 mm. Hydrogen was supplied at 100 ml min^{-1} , and the catalyst-to-feedstock ratio was fixed at 10:1. The system pressure ($0.1\text{--}3 \text{ MPa}$) was regulated via a high-pressure flow controller (HP-3050FC). Catalysts were pre-reduced at 300°C for 2 h in H_2 before each run.

Sample injection was conducted using either a manual high-pressure injector or an autosampler module capable of sequentially delivering up to 48 sealed samples without air exposure (Supplementary Figs. 13 and 14). In model experiments, styrene was vaporized at 275°C in the first stage and hydrogenolysed at 275°C and 0.4 MPa in the second stage. Catalyst stability was evaluated using 30 mg of $\text{Ru}_{\text{SA}}/\text{Co}_3\text{O}_4$ over 100 PS feeding cycles without catalyst replacement.

Product analysis was performed via an online GC–MS system equipped with a thermal conductivity detector (TCD; Agilent 8890–5977B) equipped with a 50 m HP-PONA column (0.20 mm i.d. , $0.50 \mu\text{m}$ film). The GC oven was programmed from 45°C to 320°C at $10^\circ\text{C min}^{-1}$. MS scans ranged from m/z 2 to 550, with product identification via the NIST library and external standards. Condensed liquid products were analysed offline using an Agilent 8890–7000D GC–MS with an HP-5MS column ($30 \text{ m} \times 0.25 \text{ mm} \times 0.25 \mu\text{m}$).

Scale-up validation and long-term catalyst stability

To assess the scalability and durability of $\text{Ru}_{\text{SA}}/\text{Co}_3\text{O}_4$, we conducted two scale-up experiments in fixed-bed reactors. In the first test, a standard single-stage fixed-bed reactor ($1,300 \times 600 \times 1,000 \text{ mm}$; inner tube diameter 8 mm, height 430 mm) with dual-zone temperature control was loaded with 400 mg $\text{Ru}_{\text{SA}}/\text{Co}_3\text{O}_4$, diluted in quartz sand (total bed volume 2 ml). Ethylbenzene (1.2 ml h^{-1}) was chosen as a surrogate feed to avoid styrene polymerization. After upstream vaporization, the feed entered the catalyst bed along with H_2 (50 ml min^{-1}) at 275°C and 0.4 MPa. The reaction ran continuously for 100 h with a weight hourly space velocity (WHSV) of 2.6 h^{-1} , and effluents were periodically analysed by GC–MS equipped with a flame ionization detector (FID).

In the second test, a laboratory-scale tandem reactor was used to evaluate liquid yields under realistic PS conversion conditions. Two stainless-steel tubular reactors (i.d. 8 mm) were connected in series. The first stage (400 mm, 475°C) depolymerized 200 mg of PS per run, while the second stage (350 mm, 275°C) carried out vapour-phase hydrogenolysis. Both operated at 0.4 MPa with $100 \text{ ml min}^{-1} \text{ H}_2$ flow. The catalyst-to-feed mass ratio was fixed at 10:1, with an estimated residence time of 38.2 s. Liquid products were trapped downstream using dichloromethane at -15°C and analysed via GC–MS.

DFT calculations

Spin-polarized DFT calculations were carried out using the QUICKSTEP module in CP2K (v2024.2) with the Perdew–Burke–Ernzerhof (PBE) functional and Goedecker–Teter–Hutter (GTH) pseudopotentials. A MOLOPT double- ζ valence polarized basis set and a 400 Ry plane-wave cut-off were used. Dispersion was corrected via Grimme D3 with Becke–Johnson damping. Geometry optimizations used Γ -point sampling and conventional self-consistent field (SCF) diagonalization. The Co_3O_4 (111) surface was modelled using a 4-layer slab (44 atoms). DFT + U (effective Hubbard U parameter, $U_{\text{eff}} = 5.9 \text{ eV}$ for Co, 3.3 eV for Ru) was applied to account for d-electron correlation. Density of states and charge density differences were analysed using Multiwfn, with SCPA and Gaussian broadening for partial density of states (Supplementary Methods).

Process modelling and LCA

Process simulation was performed using Aspen Plus based on experimental elemental balances. The model includes heat recovery, PS hydropyrolysis and vapour-phase hydrotreating at 275 °C, producing toluene and methane. Utility demands were estimated assuming 70% electric heating efficiency, with overall plant energy consumption benchmarked against literature, conservatively assuming heating accounts for 74% of total energy use.

A cradle-to-gate LCA was conducted per ISO 14040, with 1 kg of toluene as the functional unit. System boundaries begin at the municipal waste centre, excluding upstream plastic emissions. Plastic transport was assumed to cover 100 km. The life-cycle inventory, including key assumptions, was derived from simulation outputs and literature sources (Supplementary Notes). Methane by-product utilization was modelled via system expansion, with avoided emissions estimated using GREET 2023. GWP over a 100-year horizon (GWP100) was used as the primary environmental indicator.

TEA

A TEA was conducted using Shanghai as a representative case to evaluate the economic feasibility of the proposed process. The model focused on plastic waste streams not suitable for material recycling and assumed a feedstock cost of US\$0.05 kg⁻¹, with PS accounting for 6% of total plastic waste. Transportation costs were excluded, while sorting costs were included.

Minimum selling prices of toluene were estimated using a discounted cash flow model incorporating capital (CAPEX) and operating (OPEX) expenditures. CAPEX was calculated using the six-tenth factor rule and adjusted via the chemical engineering plant cost index. A sensitivity analysis identified feedstock cost, hydrogen price and toluene yield as key drivers. Detailed assumptions, equipment costs and parameter breakdowns are provided in Supplementary Notes.

Data evaluation

The conversion of PS was calculated as follows:

$$\text{Conversion of PS} = \frac{m_{\text{before}} - m_{\text{after}}}{m_{\text{feed}}} \times 100\%, \quad (1)$$

where m_{before} and m_{after} are the masses of the stainless-steel sample cup before and after reaction, respectively.

The yield of hydrocarbons (for example, aromatics and cycloalkanes) from the hydropyrolysis–hydrogenolysis of PS was calculated as follows:

$$\text{Yield} = \frac{m_{\text{hydrocarbons},i}}{m_{\text{polystyrene}}} \times 100\%. \quad (2)$$

Liquid products were quantified by external calibration using GC/MS. Calibration curves were established for each compound ($\geq 99.9\%$, GC grade, Sigma-Aldrich) under identical analytical conditions. All curves showed excellent linearity ($R^2 > 0.995$). Gaseous products were quantified component-by-component via GC with external standards.

Product selectivity from PS depolymerization was calculated as follows:

$$S_j = \frac{m_j}{\sum m_j} \times 100\%, \quad (3)$$

where m_j represents the mass of a specific hydrocarbon product, such as benzene, toluene, styrene, ethylbenzene or C₉, aromatics, and $\sum m_j$ denotes the total mass of all liquid hydrocarbon products.

Toluene formation rate (mmol g_{cat.}⁻¹ h⁻¹) was calculated as follows:

$$F_{\text{Toluene}} = \frac{m_{\text{Toluene}}}{92 \times m_{\text{catalyst}} \times \text{Reaction time}}, \quad (4)$$

where m_{catalyst} denotes the mass of catalyst loaded, and the reaction time corresponds to the total residence time of reactants in the reactor, calculated as follows:

$$\text{Reaction time} = \frac{V_{\text{reactor}}}{F_{\text{H}_2}}. \quad (5)$$

To ensure accurate residence time estimation, we assessed whether the tandem fixed-bed reactor operates under plug flow conditions by calculating the Péclet number (Pe) and comparing convective and diffusive timescales. A sufficiently high Pe, together with favourable dimensionless criteria, suggests near-ideal plug flow behaviour, characterized by negligible axial dispersion and effective radial mixing. Pe is defined as follows:

$$P_e = R_e \times S_c = \frac{\rho u D}{\mu} \times \frac{\mu}{\rho D_{AB}} = \frac{u D}{D_{AB}}, \quad (6)$$

where u is the superficial gas velocity (m s⁻¹), D is the characteristic length (m), ρ is the gas density (kg m⁻³), μ is the dynamic viscosity (Pa s) and D_{AB} is the molecular diffusion coefficient (m² s⁻¹).

The TON was determined as the total moles of toluene produced per mole of Ru in the catalyst during continuous operation:

$$\text{TON} = \frac{n_{\text{Toluene}}}{n_{\text{Ru}}} \quad (7)$$

where n_{Toluene} is the total amount of toluene formed, and n_{Ru} is the number of moles of Ru in the catalyst.

Data availability

All data that support the findings of this study are available from the corresponding authors upon reasonable request. Source data are provided with this paper.

Acknowledgements

This work was supported by the National Key R&D Program of China (2023YFA1506801 to D.W.) and the National Natural Science Foundation of China (52376195 to J.W. and 22171157 to D.W.). We gratefully acknowledge the 1W1B beamline of the Beijing Synchrotron Radiation Facility (BSRF) and BL14W1 beamlines at the Shanghai Synchrotron Radiation Facility (SSRF) for providing beam time. We also thank C. Guo at the Analysis Center, Department of Chemistry, Tsinghua University, for assistance with catalyst structure characterization.

Author contributions

Conceptualization: J.J. and D.W. Supervision: J.J. and D.W. Methodology: J.W., Z. Zhang, Y.Z., W.L. and D.L. Investigation: Z. Zhuang, J.W., T.H. and S.W. Theoretical calculation: Z. Zhang, D.L. and S.W. Morphology characterization: L.D. and Z. Zhang. Writing (original draft): J.W. and Z. Zhang. Writing (review and editing): D.W. and J.J.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41565-025-02069-x>.

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Peer review information *Nature Nanotechnology* thanks Guoliang Liu and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

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